

PF LAB TASKS

Course Code: C1002

Subject: Programming Fundamentals

Section: D & C

Instructor: Mr. Shayan Ali

STUDENT SUBMISSION DETAILS

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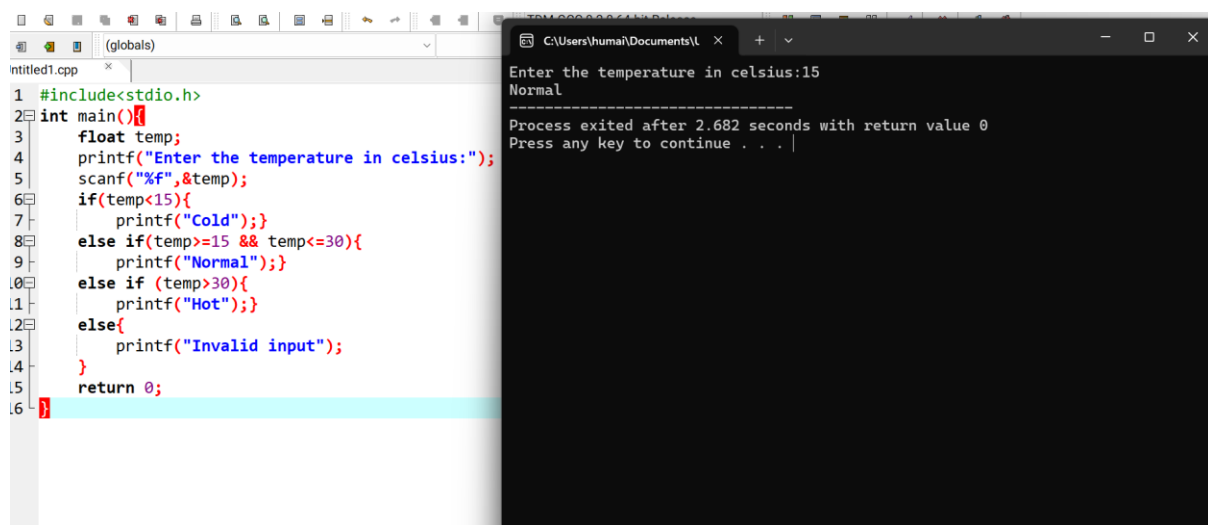
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TASK # 01 — Temperature Status (if-else)

A weather monitoring system takes the temperature in Celsius from the user. Write a C program using if-else to display “COLD” if the temperature is below 15, “NORMAL” if the temperature is between 15 and 30, and "HOT" if the temperature is above 30.



The image shows a C program in a code editor and its execution output in a terminal window. The code is as follows:

```
1 #include<stdio.h>
2 int main()
3 {
4     float temp;
5     printf("Enter the temperature in celsius:");
6     scanf("%f",&temp);
7     if(temp<15){
8         printf("Cold");
9     }
10    else if(temp>=15 && temp<=30){
11        printf("Normal");
12    }
13    else if (temp>30){
14        printf("Hot");
15    }
16    else{
17        printf("Invalid input");
18    }
19    return 0;
20 }
```

The terminal output shows the program running with the input 15, resulting in "Normal".

```
Enter the temperature in celsius:15
Normal
Process exited after 2.682 seconds with return value 0
Press any key to continue . . .
```

TASK # 02 — Mobile Data Package (if-else)

A mobile service provider checks a customer's remaining balance before activating a package. Write a C program that takes the balance as an integer and uses if-else to display "Low Balance" if the balance is below 500, "Sufficient Balance" if it is between 500 and 2000, and "Premium Balance" if it is above 2000.

The screenshot shows a C program in a text editor and its execution in a terminal window. The program uses if-else statements to check a balance and print a message. The terminal shows the user entering 3000, which results in the 'Premium Balance' message.

```

1 #include<stdio.h>
2 int main(){
3     int balance;
4     printf("Enter the balance:");
5     scanf("%d",&balance);
6     if(balance<500){
7         printf("Low Balance");
8     }
9     else if(balance>=500 && balance<=2000){
10        printf("Sufficient Balance");
11    }
12    else if (balance>2000){
13        printf("Premium Balance");
14    }
15    else{
16        printf("Invalid input");
17    }
18    return 0;
19 }

```

```

Enter the balance:3000
Premium Balance
-----
Process exited after 13.31 seconds with return value 0
Press any key to continue . . .

```

TASK # 03 — Hospital Appointment (nested if-else)

A hospital appointment system asks the patient to enter three integer values: appointment, doctorAvailable, and registrationCompleted, where 1 means yes and 0 means no. Write a C program using nested if-else statements to first check whether the patient has an appointment, then check whether the doctor is available, and finally check whether the registration is completed before displaying whether the patient can meet the doctor.

The screenshot shows a C program in a text editor and its execution in a terminal window. The program uses nested if-else statements to check three conditions: appointment, doctor availability, and registration completion. The terminal shows the user entering 1 for all three, which results in the message 'You can meet the doctor'.

```

1 #include<stdio.h>
2 int main(){
3     int appointment;
4     int doctoravailable;
5     int registrationcompleted;
6
7     printf("Is there a appointment(1=yes and 0=no):");
8     scanf("%d",&appointment);
9     printf("Is there a doctor available(1=yes and 0=no):");
10    scanf("%d",&doctoravailable);
11    printf("Is the registration completed(1=yes and 0=no):");
12    scanf("%d",&registrationcompleted);
13    if(appointment==1){
14        printf("Your appointment is confirmed\n");
15        if(doctoravailable==1){
16            printf("The doctor is available\n");
17        }
18        else{
19            printf("There is no doctor available\n");
20        }
21        if (registrationcompleted==1){
22            printf("Your registration is completed\n");
23            printf("You can meet the doctor\n");
24        }
25        else{
26            printf("Your registration is not completed\n");
27        }
28    }
29    else{
30        printf("Your appointment is not confirmed\n");
31    }
32    return 0;
33 }

```

```

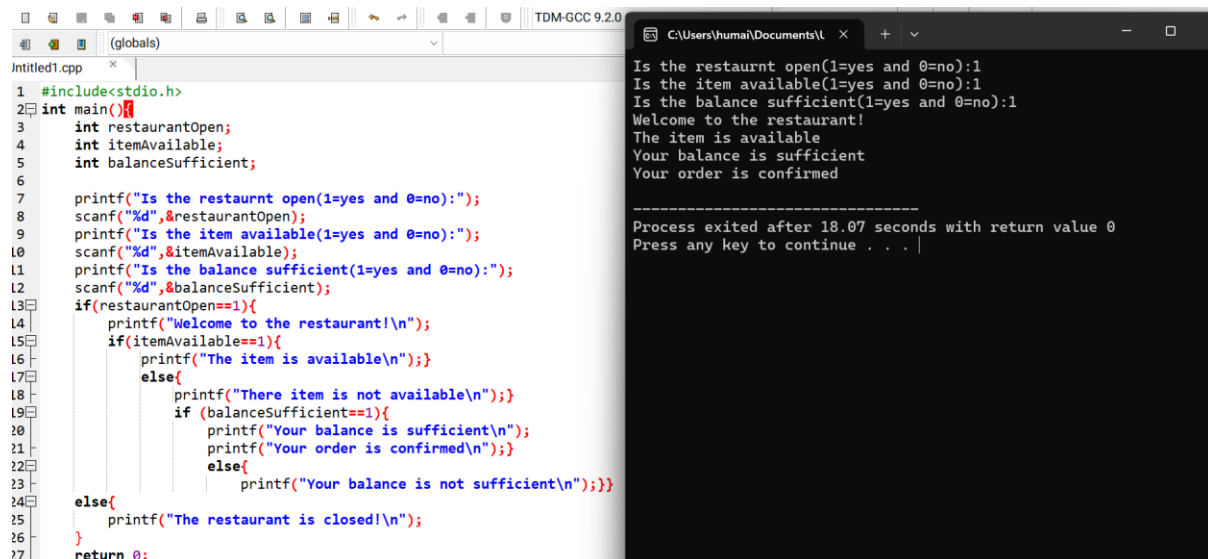
Is there a appointment(1=yes and 0=no):1
Is there a doctor available(1=yes and 0=no):1
Is the registration completed(1=yes and 0=no):1
Your appointment is confirmed
The doctor is available
Your registration is completed
You can meet the doctor
-----
Process exited after 6.052 seconds with return value 0
Press any key to continue . . .

```

TASK # 04 — Online Food Order (nested if-else)

An online food delivery system asks the customer to enter three integer values: restaurantOpen, itemAvailable, and balanceSufficient, where 1 means yes and 0 means no. Write a C program using nested if-else statements to first check whether the

restaurant is open, then whether the selected item is available, and finally whether the customer has sufficient balance before displaying the appropriate order status.



The image shows a C program in a code editor and its execution output in a terminal window. The code is a C program that checks if a restaurant is open, if an item is available, and if the balance is sufficient. It uses nested if-else statements to determine the order status.

```
1 #include<stdio.h>
2 int main()
3 {
4     int restaurantOpen;
5     int itemAvailable;
6     int balanceSufficient;
7
8     printf("Is the restaurnt open(1=yes and 0=no):");
9     scanf("%d",&restaurantOpen);
10    printf("Is the item available(1=yes and 0=no):");
11    scanf("%d",&itemAvailable);
12    printf("Is the balance sufficient(1=yes and 0=no):");
13    scanf("%d",&balanceSufficient);
14    if(restaurantOpen==1){
15        printf("Welcome to the restaurant!\n");
16        if(itemAvailable==1){
17            printf("The item is available\n");
18        }
19        else{
20            printf("There item is not available\n");
21        }
22        if(balanceSufficient==1){
23            printf("Your balance is sufficient\n");
24            printf("Your order is confirmed\n");
25        }
26        else{
27            printf("Your balance is not sufficient\n");
28        }
29    }
30    else{
31        printf("The restaurant is closed!\n");
32    }
33    return 0;
34 }
```

The terminal output shows the program's execution with the following prompts and responses:

```
Is the restaurnt open(1=yes and 0=no):1
Is the item available(1=yes and 0=no):1
Is the balance sufficient(1=yes and 0=no):1
Welcome to the restaurant!
The item is available
Your balance is sufficient
Your order is confirmed

-----
Process exited after 18.07 seconds with return value 0
Press any key to continue . . . |
```

TASK # 05 — ATM and Transaction Selection (Nested switch):

An ATM provides four operations: 1 for Balance Inquiry, 2 for Cash Withdrawal, 3 for Cash Deposit, and 4 for PIN Change. After the user selects an operation, the ATM displays a second menu related to that operation; for example, after selecting Cash Withdrawal, the user can choose 1 for Savings Account or 2 for Current Account. Write a C program using a nested switch statement to take both choices as integer inputs and display the selected operation and account type

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int operation;
6     int account;
7
8     printf("    ATM MENU    \n");
9     printf("1. Balance Inquiry\n");
10    printf("2. Cash Withdrawal\n");
11    printf("3. Cash Deposit\n");
12    printf("4. PIN Change\n");
13
14    printf("\nEnter your operation: ");
15    scanf("%d", &operation);
16
17    switch(operation)
18    {
19        case 1:
20            printf("\n    Balance Inquiry    \n");
21            printf("1. Savings Account\n");
22            printf("2. Current Account\n");
23            printf("Enter your account choice: ");
24            scanf("%d", &account);
25
26            switch(account)
27            {
28                case 1:
29                    printf("Savings Account Balance: Rs. 50,000\n");
30                    break;
31
32                case 2:
33                    printf("Current Account Balance: Rs. 80,000\n");
```

```
34                    break;
35
36                default:
37                    printf("Invalid Account Choice\n");
38            }
39            break;
40
41        case 2:
42            printf("\n    Cash Withdrawal    \n");
43            printf("1. Savings Account\n");
44            printf("2. Current Account\n");
45            printf("Enter your account choice: ");
46            scanf("%d", &account);
47
48            switch(account)
49            {
50                case 1:
51                    printf("Cash withdrawn from Savings Account.\n");
52                    break;
53
54                case 2:
55                    printf("Cash withdrawn from Current Account.\n");
56                    break;
57
58                default:
59                    printf("Invalid Account Choice\n");
60            }
61            break;
62
63
```

```
64
65    case 3:
66        printf("\n    Cash Deposit    \n");
67        printf("1. Savings Account\n");
68        printf("2. Current Account\n");
69        printf("Enter your account choice: ");
70        scanf("%d", &account);
71
72        switch(account)
73        {
74            case 1:
75                printf("Cash deposited into Savings Account.\n");
76                break;
77
78            case 2:
79                printf("Cash deposited into Current Account.\n");
80                break;
81
82            default:
83                printf("Invalid Account Choice\n");
84        }
85        break;
86
87    case 4:
88        printf("\n    PIN Change    \n");
89        printf("1. Savings Account\n");
90        printf("2. Current Account\n");
91        printf("Enter your account choice: ");
92        scanf("%d", &account);
93
94        switch(account)
```

```

84     }
85     break;
86
87
88     case 4:
89         printf("\n    PIN Change    \n");
90         printf("1. Savings Account\n");
91         printf("2. Current Account\n");
92         printf("Enter your account choice: ");
93         scanf("%d", &account);
94
95         switch(account)
96         {
97             case 1:
98                 printf("PIN changed for Savings Account.\n");
99                 break;
100
101             case 2:
102                 printf("PIN changed for Current Account.\n");
103                 break;
104
105             default:
106                 printf("Invalid Account Choice\n");
107         }
108         break;
109
110     default:
111         printf("Invalid Operation\n");
112     }
113
114     return 0;
115

```

```

84
85
86
87     ATM MENU
88     1. Balance Inquiry
89     2. Cash Withdrawal
90     3. Cash Deposit
91     4. PIN Change
92
93     Enter your operation: 3
94
95     Cash Deposit
96     1. Savings Account
97     2. Current Account
98     Enter your account choice: 2
99     Cash deposited into Current Account.
100
101     =====
102     Process exited after 8.696 seconds with return value 0
103     Press any key to continue . . . |
104
105
106
107
108
109
110
111
112
113
114
115

```